Reliable score-based confidence intervals for covariance parameters in linear mixed models

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Objectives

The objectives of our research [1] are to:

- Construct confidence intervals with correct coverage, even when parameters (including nuisance parameters) are at or near the boundary.
- Establish the asymptotically uniformly correct coverage results for a general parametric model with a Fisher information matrix growing at varying rates.
- Establish conditions for linear mixed models in particular.
- Give examples of the applicability of this theory.

Introduction

Uniformly valid confidence intervals satisfy

$$\lim_{n\to\infty} \sup_{\theta\in\mathbb{A}} |P_\theta(\theta \in C_n(\alpha)) - (1-\alpha)| = 0. \qquad (1)$$

We consider a partitioned parameter vector $\theta = (\theta^{(1)}, \theta^{(2)})$, and wish to construct confidence intervals for $\theta^{(1)}$. We do so by inverting a profile score statistic,

$$T_n(\theta^{(1)}) = S_n^{(1)}\{\theta^{(1)}, \tilde{\theta}_n^{(2)}(\theta^{(1)})\}^T \mathcal{I}_n^{(11)}(\theta^{(1)}, \tilde{\theta}_n^{(2)}(\theta^{(1)}))S_n^{(1)}\{\theta^{(1)}, \tilde{\theta}_n^{(2)}(\theta^{(1)})\}.$$

We establish distributional results that allow for the construction of confidence intervals under general conditions, including that, along a sequence of parameter values θ_n, our nuisance parameter estimate $\tilde{\theta}_n^{(2)}(\theta_n^{(1)})$ satisfies

$$\mathcal{I}_{n(22)}(\theta_n)^{1/2}(\tilde{\theta}_n^{(2)} - \theta_n^{(2)}) = \mathcal{I}_{n(22)}(\theta_n)^{-1/2}S_n^{(2)}(\theta_n) + o_p(1).$$

Such estimators include unrestricted maxima of the nuisance parameters over an enlarged parameter set, or an estimator constructed by taking one Newton-step away from any consistent estimator [2].

These score-based confidence intervals can be shown to be uniformly asymptotically valid. Our proposed confidence intervals are valid even when the true value of the parameters, including the nuisance parameters, may be at or near the boundary of the parameter set.

We also developed an `R` package, `reconf`, which implements these confidence intervals.

Linear Mixed Models

For a vector of responses $Y \in \mathbb{R}^n$ and matrices $X \in \mathbb{R}^{n\times p}$ and $Z \in \mathbb{R}^{n\times q}$, suppose

$$Y = X\beta + ZU + E, \qquad (2)$$

where the random effects vector $U \sim N_q(0, \Psi)$ and the error vector $E \sim N_n(0, \psi_r I_n)$ are independent. The covariance matrix Ψ is parameterized by the $r-1$ first elements of $\psi = [\psi_1, \dots, \psi_r]$. Specifically, $\Psi = \Sigma_{j=1}^{r-1} \psi_j H_j$ for some known $H_j \in q \times q$, $j \in \{1, \dots, r-1\}$. Both matrices $\Psi(\psi)$ and $\Sigma(\psi)$ are parametrized by ψ. Assuming X and Z are non-stochastic, (2) implies

$$Y \sim N_n(X\beta, \Sigma), \quad \Sigma = Z\Psi Z^T + \psi_r I_n. \qquad (3)$$

Variance components models are important special cases of (3) which result from taking $Z = I_n$ and H_j symmetric and positive semi-definite, $j \in \{1, \dots, r-1\}$. Indeed, doing so gives $\Sigma = \Sigma_{j=1}^{r-1}\psi_j H_j + \psi_r I_n$ with $\psi_j \geq 0$, $j \in \{1, \dots, r\}$.

Coverage Simulations

Consider a model with correlated random effects

$$Y_{ij} = X_{ij}^T\beta + Z_{ij}^T U_i + E_{ij}$$

$U_i \sim N(0, \Psi_1)$, with

$$\Psi_1 = \begin{bmatrix} \psi_1 & \psi_2 \\ \psi_2 & \psi_3 \end{bmatrix}$$

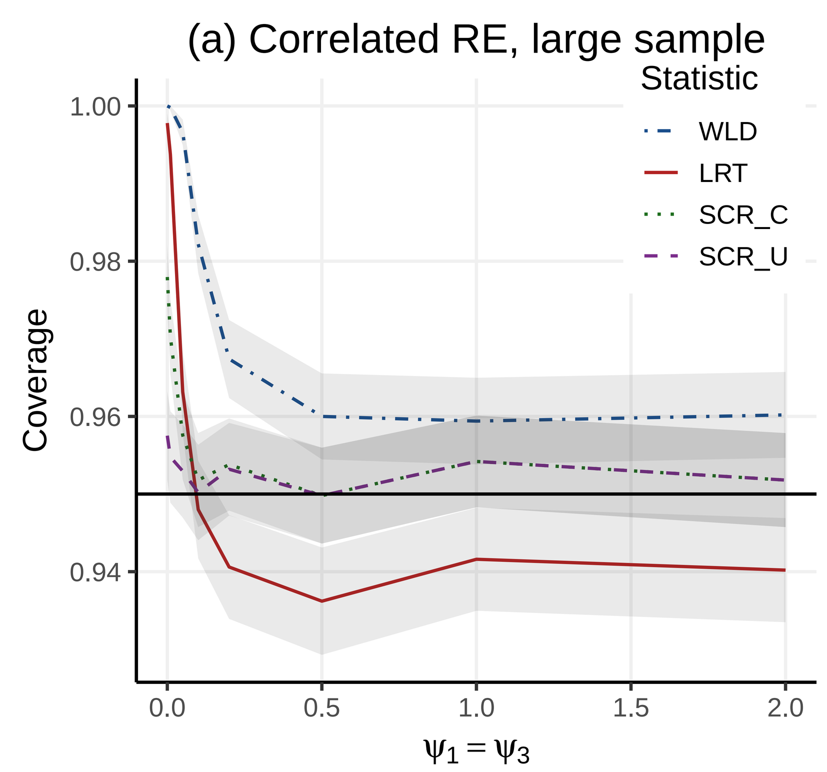

Figure 1: Coverage probabilities with independent clusters and correlated random effects, for different values of the random effect variances $\psi_1 = \psi_3$. Parameter of interest is the correlation ψ_2. The settings are (n_1, n_2, p) equal to $(20, 200, 1)$

Results

Under general assumptions, the profile score statistic,

$$T_n(\theta_n^{(1)}) \overset{\theta_n}{\rightsquigarrow} \chi^2_{d_1}, \qquad (4)$$

the most important assumption being For every n,

$$S_n^{(1)}(\theta_n^{(1)}; \tilde{\theta}_n^{(2)}) = S_n^{(1)}(\theta_n) - \mathcal{I}_{n(12)}(\theta_n)(\tilde{\theta}_n^{(2)} - \theta_n^{(2)}) + R_n(\theta_n, \tilde{\theta}_n^{(2)}),$$

where $\mathcal{I}_{n(11)}(\theta_n)^{-1/2}R_n(\theta_n, \tilde{\theta}_n^{(2)}) = o_p(1)$. This can be shown by establishing uniform convergence in probability of the observed to the expected Fisher information. General conditions for linear mixed models include the condition that for each $i, j = 1, \cdots, r$,

$$\sup_{\psi\in\left\{\theta: ||\mathcal{I}_n(\theta_n)^{1/2}(\theta-\theta_n)|| \leq M\right\}} ||M_{n,ij}(\psi)|| = o\left(\frac{\lambda_{min}(\mathcal{I}_n)}{n}\right)$$

where

$$M_{n,ij} = \Sigma(\psi_n)^{1/2}\Sigma(\psi)^{-1/2}A_i(\psi)A_j(\psi)\Sigma(\psi)^{-1/2}\Sigma(\psi_n)^{1/2} - A_i(\psi_n)A_j(\psi_n)$$

and

$$A_{n,i}(\psi) := \Sigma_n(\psi)^{-1/2} Z H_i Z^T \Sigma_n(\psi)^{-1/2}$$

Application: SNP Data

Single nucleotide polymorphisms (SNPs) are locations of the genome at which a single nucleotide can vary between populations. One can stratify the SNPs into groups by minor allele frequency (MAF). We consider 2 bins, and hence a mixed effects model with 2 variance components.

$$y = X\beta + g_1 + g_2 + \epsilon$$

where $g_i \sim N(0, \sigma_i^2 K_i)$, $i = 1, 2$. K_i is the "genetic relationship matrix", derived from the genotypes of the individuals, for MAF bin i.

The table below shows confidence intervals for the three covariance parameters $(\sigma_1^2, \sigma_2^2, \sigma_\epsilon^2)$ constructed using `lme4` and `reconf`. The data we use is `DOGatti_2014` from `qtl2` package, which contains mouse genotypes and white blood cell count (the response). Fixed effects are sex, number of generations of outbreeding, and birth date of the mouse.

Profile Score CIs computed via a one-step score-based inversion using `reconf::ci_all_lmer`. Profile Likelihood CIs obtained by inverting the likelihood ratio statistic via `lme4::profile`.

Table 1: Variance component estimates and 95% confidence intervals from two methods applied to the mouse SNP data ($n = 200$ individuals, 598 low-MAF and 5815 high-MAF markers).

		95% Confidence Interval	
Component	MLE	Profile Score	Profile Likelihood
σ_1^2	0.000	(0.000, 1.187)	(0.000, 1.227)
σ_2^2	1.708	(0.000, 4.788)	(0.000, 3.925)
σ_ε^2	3.589	(1.821, 5.821)	(1.863, 5.352)

Conclusion

Our paper presents a general method for constructing confidence intervals by inverting a profile score statistic, and this confidence interval has asymptotically correct coverage probability in situations where alternative confidence intervals, such as those constructed using the Likelihood Ratio and Wald statistics, do not. In applications, such as the analysis of SNP Data, highly correlated random effect estimates lead to appreciable differences in coverage probability, and illustrate the benefit of using this procedure.

References

[1] Matias Shedden and Karl Oskar Ekvall.
Reliable score-based confidence intervals for covariance parameters in linear mixed models.
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[2] Philipp Ketz.
Subvector inference when the true parameter vector may be near or at the boundary.
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